

16x16 2D Systolic Array Hardware Accelerator

Vivado 2025.1 ML Edition: Complete Simulation, Synthesis & Hardware Guide

Target Hardware: TUL PYNQ-Z2 (Zynq-7000 xc7z020clg400-1)
Baseline Board: Digilent Basys 3 (Artix-7 xc7a35tcbg236-1)

Array Scale: 16x16 Mesh (256 Processing Elements)
Precision: INT8 Inputs (Signed) -> INT16 Accumulation
Core Latency: 48 Clock Cycles (384 ns @ 125 MHz)

Important Note on Testbench Naming (Step 3 vs 3x3):

The file `tb_step3_skew_buffer.sv` represents **Development Step 3** in the incremental roadmap (which specifically verified the Skew Buffer module that introduces diagonal time-staggering). It is **not** a 3x3 matrix. The skew buffer is fully parameterized by N and feeds the full 16x16 grid. All testbenches are located directly in `boards/pynq_z2/sim/` for 1-click access in Vivado.

1. Clean Directory Structure (Zero Global Clutter)

All testbenches have been consolidated directly inside the board directory (`boards/pynq_z2/sim/`) so that when you open Vivado, all files appear directly under your project hierarchy without searching through external folders:

Category	Folder Location	Included Files	Role in Vivado
Board Implementation	<code>boards/pynq_z2/board_implementation/</code>	<code>pynq_z2_demo_16x16_top.sv</code> <code>pynq_z2_constraints.xdc</code>	Top module for hardware bitstream synthesis. Contains ROM matrices, switch decoder, debouncers, status LEDs, and UART output.
RTL Design Sources	<code>boards/pynq_z2/src/</code> (or <code>rtl/</code>)	<code>systolic_pkg.sv</code> , <code>processing_element.sv</code> , <code>skew_buffer.sv</code> , <code>systolic_array.sv</code> , <code>controller.sv</code> , <code>systolic_top.sv</code>	Synthesizable hardware accelerator modules. Top module is <code>systolic_top.sv</code> .
Unit Testbenches	<code>boards/pynq_z2/sim/</code>	<code>tb_step1_pe.sv</code> to <code>tb_step5_systolic_8x8.sv</code> <code>tb_step6_systolic_16x16.sv</code>	Incremental self-checking testbenches. <code>tb_step6_systolic_16x16</code> validates the full 16x16 array (Identity, Scaled, Dense GEMM).
OOP Testbench Suite	<code>boards/pynq_z2/sim/</code>	<code>tb_matrix_top.sv</code> , <code>MatrixGenerator.sv</code> , <code>MatrixScoreboard.sv</code> , <code>MatrixDriver.sv</code> , <code>MatrixMonitor.sv</code> , <code>MatrixTransaction.sv</code>	Enterprise randomized UVM-style verification running 30 randomized transactions (7,680 individual checks, 0 mismatches).

2. Vivado 2025.1 Project Setup (GUI Mode)

Open Vivado 2025.1 ML Edition and configure your project in 4 simple steps:

Step	Action in Vivado GUI	Exact Selection / File Path
1	Create New Project	Click Create Project -> Project Name: <code>systolic_16x16</code> -> Select RTL Project .
2	Select Target FPGA Part	In Default Part, select: xc7z020clg400-1 (PYNQ-Z2 Zynq-7000) or xc7a35tcbg236-1 (Basys 3).

3	Add Design Sources & Constraints	• Add Design Sources: Add all files in boards/pynq_z2/src/ and boards/pynq_z2/board_implementation/pynq_z2_demo_16x16_top.sv. • Add Constraints: Add boards/pynq_z2/board_implementation/pynq_z2_constraints.xdc.
4	Add Simulation Sources	Add Simulation Sources: Add all .sv files from boards/pynq_z2/sim/ .

3. How to Set Top Module in Vivado

In Vivado's **Sources** tab, you can select which operation to run simply by right-clicking a file and choosing **Set as Top**:

Desired Operation	Target Top File	Top Module Name	Step-by-Step Action in Vivado GUI
16x16 Unit Simulation (3 Fixed Verification Tests)	boards/pynq_z2/sim/ tb_step6_systolic_16x16.sv	tb_step6_systolic_16x16	1. In Sources tab, expand Simulation Sources. 2. Right-click tb_step6_systolic_16x16.sv -> select Set as Top. 3. In Flow Navigator, click Run Simulation -> Run Behavioral Simulation.
OOP Verification (30 Randomized Transactions)	boards/pynq_z2/sim/ tb_matrix_top.sv	tb_matrix_top	1. In Simulation Sources, right-click tb_matrix_top.sv -> Set as Top. 2. Click Run Behavioral Simulation. 3. Observe Scoreboard print 30 passed transactions (7,680 checks).
Intermediate Step Tests (PE, 2x2, Skew, 4x4, 8x8)	boards/pynq_z2/sim/ tb_step[1..5]_*.sv	tb_step[1..5]_*	Right-click any step testbench (e.g. tb_step3_skew_buffer.sv) -> Set as Top -> Run Simulation to verify intermediate building blocks.
PYNQ-Z2 Bitstream (Hardware FPGA Build)	boards/pynq_z2/board_implementation/ pynq_z2_demo_16x16_top.sv	pynq_z2_demo_16x16_top	1. In Sources tab, expand Design Sources. 2. Right-click pynq_z2_demo_16x16_top.sv -> Set as Top. 3. In Flow Navigator, click Generate Bitstream.

4. Waveform & Timing Interpretation

When running behavioral simulation in Vivado XSim, the 16x16 accelerator exhibits strict deterministic timing:

Key Waveform Signals to Monitor:

- **clk / sysclk** : 100 MHz clock (10 ns period) in simulation, 125 MHz on PYNQ-Z2.
- **rst** : Active-high synchronous reset pulse (held high for 2 clock cycles).
- **start** : 1-clock-cycle pulse triggering FSM transition from IDLE to LOAD.
- **load_en** : High for exactly 1 cycle while parallel matrices are latched into skew buffers.
- **shift_en & array_en** : High for $(3N - 1) = 47$ clock cycles while diagonal wavefront feeds the 256 PEs.
- **done** : Asserted exactly at cycle 48 (500 ns @ 100 MHz / 384 ns @ 125 MHz).
- **result[15:0][15:0]** : Holds the complete 16x16 signed matrix C when done is asserted.

5. PYNQ-Z2 Hardware Demonstration Guide

Once the bitstream is programmed via Vivado Hardware Manager, operate the board as follows:

Control / Indicator	Pin / Color	Hardware Function & Meaning
Slide Switches sw[1:0]	Pins M19, M20	00 : Matrix A * Identity ($C = A$) 01 : Matrix A * 2*Identity ($C = 2A$) 10 : Dense Signed INT8 GEMM 11 : All-Ones Test ($C[i][j] = 16$)
btn[1] (START)	Pin D20	Press once to initiate 16x16 matrix multiplication across all 256 PEs.

btn[0] (RESET)	Pin D19	Synchronous reset clears internal registers and latches.
led[0] (Heartbeat)	Pin R14	Smoothly blinks at ~1.86 Hz to verify the 125 MHz clock and FPGA logic are active.
led[1] (Busy)	Pin P14	High during the 48 cycles of active systolic compute.
led[2] (DONE)	Pin N16	Lights up solid when the full 16x16 matrix is calculated.
led[3] (PASS)	Pin M14	256/256 Hardware Verification PASS indicator. Lights up solid when all elements match expected golden results.
rgbled4 (Core State)	Pins M15, T16, Q15	Blue: IDLE / Ready Amber: COMPUTING Vibrant Green: DONE & 100% VERIFIED
UART Serial Output	Pin Y18 (PMOD A1)	Connect USB-UART adapter at 115200 Baud, 8N1 . Prints 16x16 GEMM: PASS upon completion.